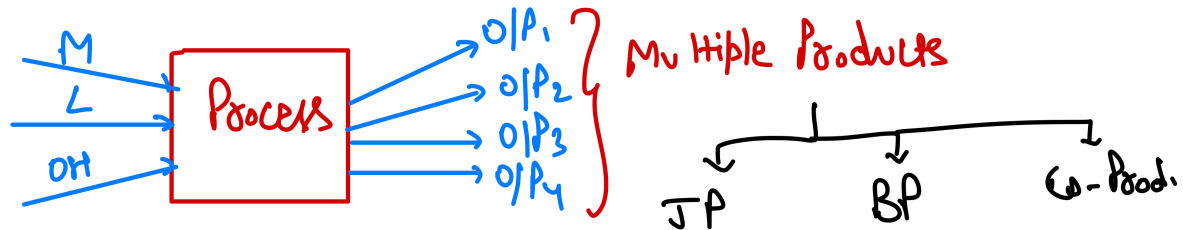
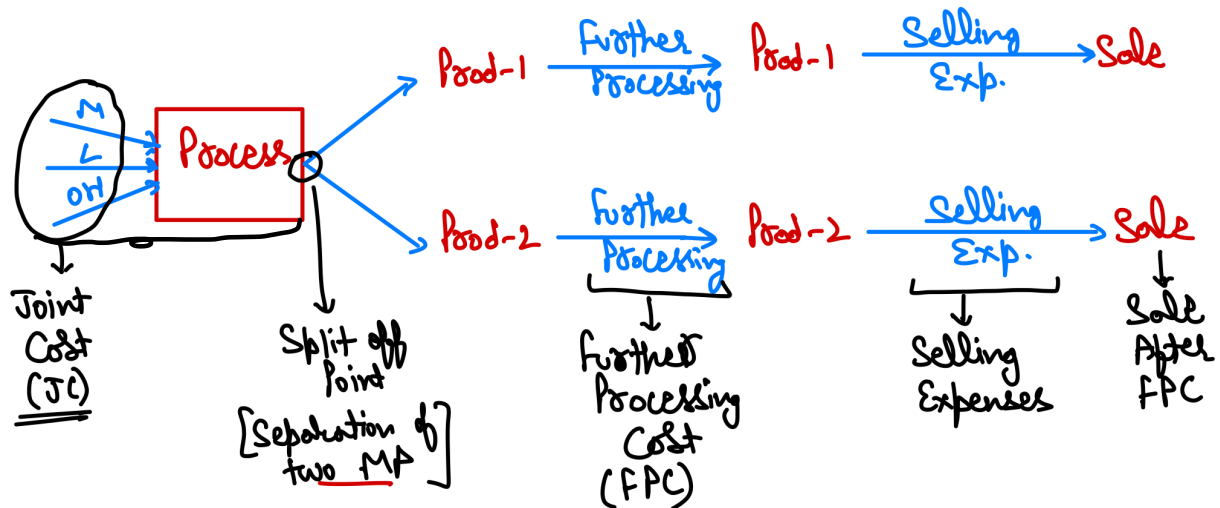


JOINT & BY-PRODUCT

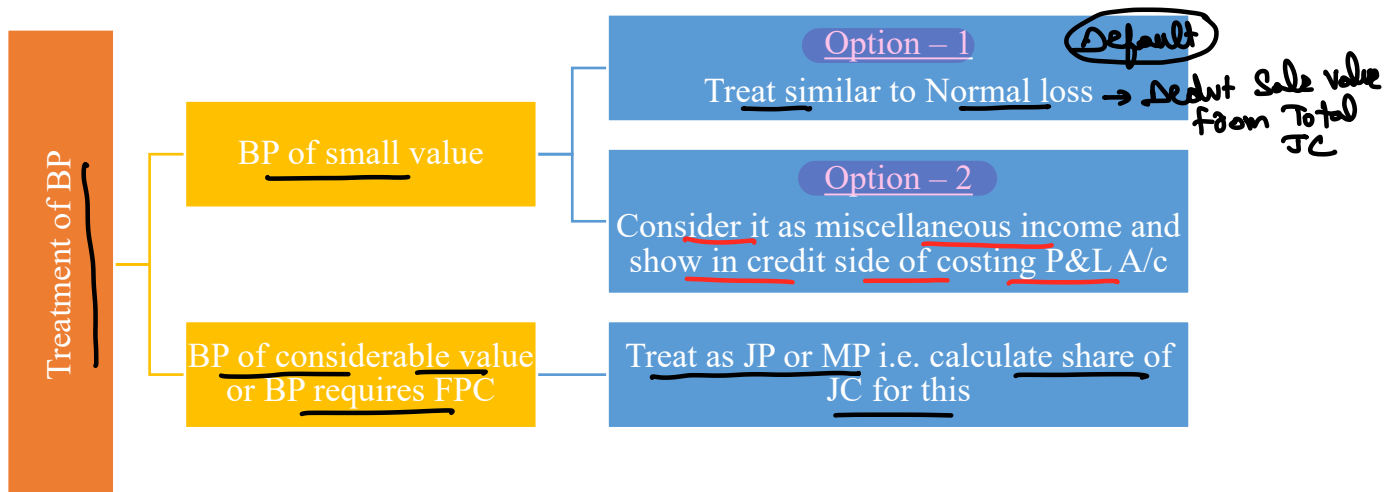
1. Multiple Output



2. Meaning of Basic Terms



3. Treatment of By-Product



4. Methods of Apportionment of Joint Cost (JC)

(A) Reverse Cost Method

Particulars	Product A	Product B
Sale value →	-	-
Less: Profit (Given in Ques.) →	-	-
Total Cost →	-	-
Less: Selling expenses →	-	-
CoP / Work Cost / Fac. Cost →	-	-
Less: Further Processing cost →	-	-
Share of Joint Cost →	✓	✓

① JC to be distributed on the basis of unit Produced

② Profit to be calculated on unit Sold

③ Production = ✓
 (⇔) Op. Stock = ✓
 (⇔) Cl. Stock = ✓
 COGS = ✓
 (⇔) Pbt. = ✓
 Sale = ✓

④ FIFO Method

If total joint cost is not matching with value given in question then distribute actual joint cost given in question in the share of joint cost calculated from above statement.

If data of profit is given for only one product then prepare above table for one product after that share of JC of other product = Total JC – Share of JC from above table for one product.

(B) Physical Unit Method – Distribute on the basis of physical units produced

(C) Sale value at split off method – Distribute on the basis of sale value at split off of units produced.

(D) Sale value after FPC method – Distribute on the basis of sale value after FPC of units produced.

(E) Net Realizable Value (NRV) Method – Distribute on the basis of NRV

NRV = Sale after FPC – Selling Expenses – FPC – Est. Profit (if Given in Ques.)

(F) Contribution Margin Method

- Distribute variable cost on the basis of units produced- Distributed fixed cost on the basis of contribution- If any product has zero or negative contribution than FC will not be distributed for that product.

TC → VC → No. of units
 TC → FC → Contribution (Sale - VC)

(G) Average Cost Method

Average cost = $\frac{\text{Total Joint Cost} \checkmark}{\text{Total Units at Separation Point}}$

5. Decision Regarding Further Processing

Compare Incremental Revenue with Incremental Cost

If Incremental revenue > Incremental cost - Yes further process

If Incremental revenue < Incremental cost - Not to further process

Incremental revenue = Sale after FPC - Sale at split off

Incremental cost = FPC + Increase in any other cost - Decrease in any other cost

option - A
Sale at split off
 (-) Sell. Exp.
 (-) x
 (-) Sh. of JC

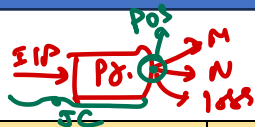
Pft.

option - B
Sale after FPC
 (-) Sell. Exp.
 (-) FPC
 (-) Sh. of JC

Pft.

Incremental (B-A)
Inc. Sale
 (-) Inc. S. Exp.
 (-) FPC

Inc. Profit



$$\text{Total JC} = 5.40\text{l} + 2.25\text{l} = 7.65\text{l}$$

Question – 1

A company's plant processes 6,750 units of a raw material in a month to produce two products 'M' and 'N'. The process yield is as under:

Product M → 80%

Product N → 12%

Process loss → 8%

$$(2.15\text{l} \times 66\% \text{ in } 100:80) \text{ Lab. Cost}$$

$$(7.65\text{l} - 1.485\text{l}) = 6.165\text{l}$$

$$\text{in } 80:12$$

	M	N
	82,500	66,000
	5,36,087	80,413

The cost of raw material is ₹ 80 per unit. $\times 6750 = 5.40\text{l}$

Processing cost is ₹ 2,25,000 of which labour cost is accounted for 66%. Labour is chargeable to products 'M' and 'N' in the ratio of 100:80.

Prepare a comprehensive cost statement for each product showing:

- Apportionment of joint cost among products 'M' and 'N' and
- Total cost of the products 'M' and 'N'.

Solution

Total joint cost = Raw material cost + Processing cost = $(6,750 \times 80) + 2,25,000 = ₹ 7,65,000$

Total labour cost = $2,25,000 \times 66\% = ₹ 1,48,500$

Joint cost other than labour cost = $7,65,000 - 1,48,500 = ₹ 6,16,500$

Statement of Joint Cost Apportionment

Particulars	Product M	Product N
Labour Cost → (1,48,500 in 100:80)	82,500 ✓	66,000 ✓
Cost other than labour cost (6,16,500 in 80:12)	✓ 5,36,087	80,413
Share of Joint Cost	6,18,587	1,46,413

Statement of Total Cost

Particulars	Product M	Product N
Raw material cost → (5,40,000 in 80:12)	4,69,565	70,435
Labour Cost → (1,48,500 in 100:80)	82,500	66,000
Other Processing Cost → (76,500 in 80:12)	66,522	9,978
Share of Joint Cost	6,18,587	1,46,413

$$\text{Real. value of B} = \frac{1}{125\%} \times 1000 = 800$$

$$\text{Net JC} = 90800 - 800 = 90000$$

Question – 2

Three products X, Y and Z alongwith a byproduct B are obtained again in a crude state which require further processing at a cost of ₹5 for X; ₹4 for Y; and ₹2.50 for Z per unit before sale. The byproduct is however saleable as such to a nearby factory. The selling prices for the three main products and byproduct, assuming they should yield a net margin of 25% of cost, are fixed at ₹13.75, ₹8.75 and ₹7.50 and ₹1.00 respectively – all per unit quantity sold.

During a period, the joint input cost including the material cost was ₹90,800 and the respective outputs were:

X	→ 8,000 units
Y	→ 6,000 units
Z	→ 4,000 units
B	→ 1,000 units

	$\frac{13.75}{125\%} = 11$	$\frac{8.75}{125\%} = 7$	$\frac{7.50}{125\%} = 6$
TC			
(-) FPC			
JC P.U.	6	3	3.5
JC	48000	18000	14000
Actual JC			

By-product should be credited to joint cost and only the net joint costs are to be allocated to the main products. Calculate the joint cost per unit of each product and the margin available as a percentage on cost.

Solution**Computation of share of joint costs**

Particulars	X (₹)	Y (₹)	Z (₹)
Selling price →	13.75	8.75	7.50
Less: Profit @ 25% on cost or 20% on sales →	2.75	1.75	1.50
Cost of sales →	11.00	7.00	6.00
Less: Post split off cost →	5.00	4.00	2.50
Joint cost per unit →	6.00	3.00	3.50
Output (units) →	8,000	6,000	4,000
Total cost	48,000	18,000	14,000
Actual share of net joint cost (A) → (90,000 in 48:18:14)	54,000	20,250	15,750
Output (units) (B) →	8,000	6,000	4,000
Joint cost per unit (A ÷ B) →	6.75	3.38	3.94

Statement of Profit

Particulars	X	Y	Z
Joint cost per unit →	6.75	3.38	3.94
Post split off cost →	5.00	4.00	2.50
Total cost →	11.75	7.38	6.44
Selling price →	13.75	8.75	7.50
Margin →	2.00	1.37	1.06
Margin % on cost	17.02%	18.56%	16.46%

$$24,000 = 10,000 + 8,000 + \text{Total Selling Expenses} + 5,600$$

$$\text{Total Selling Expenses} = ₹ 400$$

$$\text{Ratio of sales} = 16,000 : 8,000 = 2:1$$

$$\text{Selling expenses of P} = 400 \times (2/3) = ₹ 267$$

$$\text{Selling expenses of Q} = 400 \times (1/3) = ₹ 133$$

Statement of Cost of Production

Particulars	Product P	Product Q
Joint Cost:		
Material (5,000 in 6,733:3,267) }	3,367	1,633
Labour (3,000 in 6,733:3,267) }	2,020	980
Overheads (2,000 in 6,733:3,267) }	1,347	653
Further Processing Cost:		
Material }	3,000	1,500
Labour }	1,400	1,000
Overheads }	600	500
Cost of Production	11,733	6,267

Question – 4

The SK Oil company purchase crude vegetable oil. It does refining of the same. The refining process results in four products at the split off point: M, N, O & P.

Product O is fully processed at the split off point. Product M, N, & P can be individually further refined into 'Super M', 'Super N' and 'Super P'. In the most recent month (October), the output at split off point was:

Product M	→ 3,00,000 gallons
Product N	→ 1,00,000 gallons
Product O	→ 50,000 gallons
Product P	→ 50,000 gallons

The joint cost of the purchasing the crude vegetable oil and Processing it were ₹ 40,00,000. SK had no beginning or ending inventories. Sales of Product O in October were ₹ 20,00,000. Total output of products M, N and P was further refined and then sold. Data related to October are as follows:

Product	Further processing cost to make super products (FPC)	Sales after FPC	NPV
Super M	₹ 80,00,000	₹ 1,20,00,000	400
Super N	₹ 32,00,000	₹ 40,00,000	80
Super P	₹ 36,00,000	₹ 48,00,000	120

SK had the option of selling products M, N and P at the split off point. This alternative would have yielded the following sales for the October, production:

		<u>M</u>	<u>N</u>	<u>P</u>
Product M → ₹ 20,00,000	Inc. Rev. → FPC	100ℓ (80ℓ)	28ℓ (32ℓ)	20ℓ (36ℓ)
Product N → ₹ 12,00,000	Inc. Pft.	20ℓ	(4ℓ)	(16ℓ)
Product P → ₹ 28,00,000	Further Pr.	Yes	No	No

You are required to answer:

(a) How the joint cost of ₹ 40,00,000 would be allocated between each product under the following methods

- (i) Sales value at split off point → M 20ℓ : N 12ℓ : O 20ℓ : P 28ℓ
 (ii) Physical output (gallon) → 3ℓ : 1ℓ : 0.5ℓ : 0.5ℓ
 (iii) Estimated net realizable value → 40ℓ : 8ℓ : 20ℓ : 12ℓ

(b) Could SK have increased its October operating profits by making different decisions about the further refining of product M, N or P? Show the effect of any change you recommend on operating profits.

Solution

(a) (i) Statement of Cost (₹ in lakhs)

Particulars	Product M	Product N	Product O	Product P
Sale value at split off	20	12	20	28
Share of Joint Cost (40 in 20:12:20:28)	10	6	10	14

(a) (ii) Statement of Cost (₹ in lakhs)

Particulars	Product M	Product N	Product O	Product P
Physical Output	3	1	0.50	0.50
Share of Joint Cost (40 in 3:1:0.5:0.5)	24	8	4	4

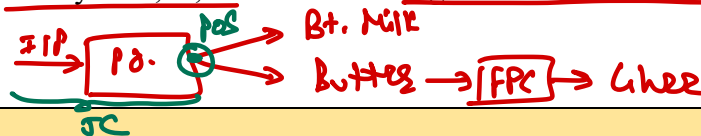
(a) (iii) Statement of Cost (₹ in lakhs)

Particulars	Product M	Product N	Product O	Product P
Sale value after FPC	120	40	20	48
Less: Further Processing cost	80	32	-	36
Less: Selling expenses	-	-	-	-
Net Realizable Value	40	8	20	12
Share of Joint Cost (40 in 40:8:20:12)	20	4	10	6

(b) Statement of Incremental Profit/(Loss) (₹ in lakhs)

Particulars	Product M	Product N	Product P
Sales after FPC	120	40	48
Less: Sale at Split off	20	12	28
Incremental Sales	100	28	20
Less: Further Processing Cost	80	32	36
Incremental Profit/(Loss)	20	(4)	(16)

Thus, it is recommended that Product M should be further processed into Super M and Product N & P should be sold at split off point without any further processing. With this suggestion, company's operating profit will rise by ₹ 20,00,000 as the existing losses of ₹ 4,00,000 and ₹ 16,00,000 will get cut down.



$$JC = 12 + (50000 \times 100) = ₹. 512$$

Question – 5

SK Ltd. is engaged in the production of Buttermilk, Butter and Ghee. It purchases processed cream and let it through the process of churning until it separates into buttermilk and butter. For the month of May, 2021, SK Ltd. purchased 50 kilolitre processed cream @ ₹ 100 per 1,000 ml. Conversion cost of ₹ 1,00,000 were incurred up-to the split off point, where two saleable products were produced i.e. buttermilk and butter. Butter can be further processed into Ghee.

The May, 2021 production and sales information is as follows:

Products	Production (in Kilolitre/tonne)	Sales Quantity (in Kilolitre/tonne)	Selling price per litre/kg (₹)
Buttermilk	28	28	30
Butter	20	-	-
Ghee	16	16	480

$$\begin{aligned}
 &76.80 - 720 \\
 &= 4.80 \\
 &(1.20) \\
 &3.60 \\
 &\text{Yes FPC}
 \end{aligned}$$

All 20 tonne of butter were further processed at an incremental cost of ₹ 1,20,000 to yield 16 Kilolitre of Ghee. There was no opening or closing inventories of buttermilk, butter or ghee in May, 2021.

Required:

- Show how joint cost would be apportioned between Buttermilk and Butter under estimated net realizable value method.
- MP Ltd. offers to purchase 20 tonne of butter in June at ₹ 360 per kg. In case SK Ltd. accepts this offer, no Ghee would be produced in June. Suggest whether SK Ltd. shall accept the offer affecting its operating income or further process butter to make Ghee itself?

Solution

- Total Joint Cost = Processed cream cost + conversion cost

$$= (50 \times 1,000 \times 100) + 1,00,000 = ₹ 51,00,000$$

Statement of Joint Cost

Particulars	Buttermilk Amount (₹)	Butter Amount (₹)
Sales Value	$30 \times 28 \times 1,000 = 8,40,000$	$16 \times 1,000 \times 480 = 76,80,000$
Less: Post split-off cost	-	(1,20,000)
Net Realizable Value	8,40,000	75,60,000
Apportionment of Joint Cost of ₹ 51,00,000 in ratio of 1:9	5,10,000	45,90,000

(ii) Statement of Incremental Profit or Loss

Particulars	(₹)
Revenue from Ghee	$16 \times 1,000 \times 480 = 76,80,000$ ✓
(-) Revenue from Butter	$20 \times 1,000 \times 360 = 72,00,000$ ✓
Incremental Revenue	→ 4,80,000
(-) Further processing cost	→ 1,20,000
Incremental Profit	3,60,000

The operating income of SK Ltd. will be reduced by ₹ 3,60,000 in June if it sells 20 tonne of Butter to MP Ltd., instead of further processing of Butter into Ghee for sale. Thus, SK Ltd. is advised not to accept the offer and further process butter to make Ghee itself.

Question – 6

A company produces two products A and B, through a joint production process. The total joint production cost incurred is as under:

Material	₹ 20,000
Labour	₹ 10,000
Variable overheads	₹ 6,000
Fixed overheads	₹ 24,000

VC = 60,000
FC = 24,000
JC = 84,000

	A	B
Sales	40,000	60,000
(-) VC (36,000 in 2:4)	12,000	24,000
Cont.	28,000	36,000
FC (24,000 in 28:36)	10,500	13,500
Share of JC	22,500	37,500
(VC + FC)		

Product A and B can be sold for ₹ 20 per unit and ₹ 15 per unit respectively at split off point. The produced quantities are Product A-2,000 units and Product B – 4,000 units.

- (i) You are required to calculate the joint production cost allocation for each product using the:
- Physical unit method. A : B → 2 : 4
 - Contribution margin method.
- (ii) Product B can be further processed by incurring expenditure of ₹ 12,000. Loss in further processing is 2%. It can be sold @ ₹ 18 per unit. Explain the impact on profitability if Product B is further processed.

Sale after FPC = $(4000 \times 18) = 72,000$
 (-) Sale at split = 60,000
 IP = 12,000
 (-) FPC = 12,000
 Inc. Pft. (1440)
 Not to FPC

Solution

- (i) (a) Total Joint Cost = 20,000 + 10,000 + 6,000 + 24,000 = ₹ 60,000

Share of Joint cost of Product A = $60,000 \times \frac{2,000}{6,000} = ₹ 20,000$

Share of Joint cost of Product B = $60,000 \times \frac{4,000}{6,000} = ₹ 40,000$

- (b) Total Variable cost = 20,000 + 10,000 + 6,000 = ₹ 36,000
 Total Fixed cost = ₹ 24,000

Statement of Cost

Particulars	Product A	Product B
Sales Value (A)	$2,000 \times 20 = 40,000$	$4,000 \times 15 = 60,000$
Variable Cost (B) [36,000 in quantity ratio 2:4]	12,000	24,000

Contribution (A – B)	28,000	36,000
Fixed cost (C) [24,000 in 28:36]	10,500	13,500
Share of Joint Cost (B + C)	22,500	37,500

(ii)

Statement of Profit of Product B

Particulars	Amount
Units produced and sold 98% of 4,000 units	3,920
Selling price per unit	18
Sales value after FPC	70,560
Sales value at split off	60,000
Incremental revenue	10,560
Less: Further processing cost	12,000
Incremental loss after further processing	(1,440)

Thus, there is a net loss of ₹ 1,440 if the Product B is further processed and sold.



Question – 7

ABC Company produces a Product 'X' that passes through three processes: R, S and T. Three types of raw materials, viz., J, K and L are used in the ratio of 40:40:20 in process R. The output of each process is transferred to next process. Process loss is 10% of total input in each process. At the stage of output in process T, a by-product 'Z' is emerging and the ratio of the main product 'X' to the by-product 'Z' is 80:20. The selling price of product 'X' is ₹ 60 per kg.

SV

$$\frac{14580 \times 60}{80} = 87480$$

$$\frac{18225 \times 20}{20} \times 30 = 109350$$

The company produced 14,580 kgs of product 'X'.

Material price: Material J @ ₹ 15 per kg; Material K @ ₹ 9 per kg.

Material L @ ₹ 7 per kg. Process costs are as follows:

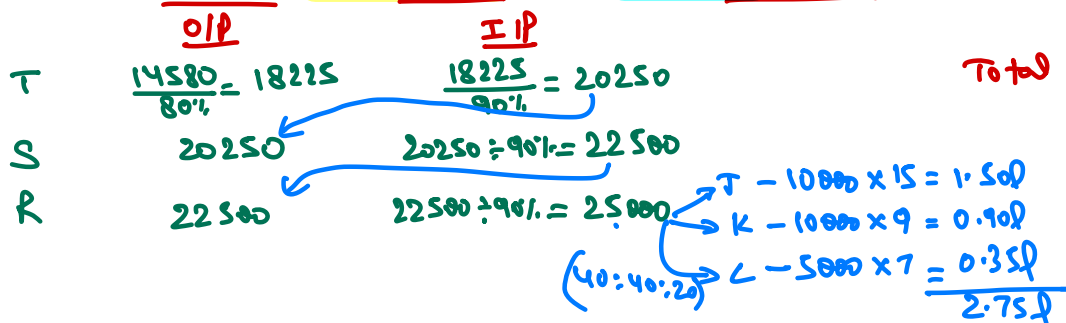
Process	Variable cost per kg (₹)	Fixed cost of input (₹)
R	$25000 \times 5.00 = 1.250$	+ 42,000 = 1.670
S	$22500 \times 4.50 = 101250$	+ 5,000 = 106250
T	$20250 \times 3.40 = 68850$	+ 4,800 = 73650

The by-product 'Z' cannot be processed further and can be sold at ₹ 30 per kg at the split-off stage.

There is no realizable value of process losses at any stage.

Required:

Present a statement showing the apportionment of joint costs on the basis of the sales value of product 'X' and by-product 'Z' at the split-off point and the profitability of product 'X' and by-product 'Z'.



$$\text{Total JC} = 2.750 + 346900 = 621900$$

Solution**Working Note:**

(1) Let total raw material in Process R of raw material R be 100

Process	Input	Loss	Output
R	100	$100 \times 10\% = 10$	90
S	90	$90 \times 10\% = 9$	81
T	81	$81 \times 10\% = 8.10$	72.90

Thus, for input of 100 units in process R, output of 72.90 units is obtained from process T.

Actual output of X = 14,580 units

80% of output of Process T = 14,580 units

Output of process T = $14,580 \div 80\% = 18,225$

Input of process R = $\frac{18,225}{72.9} \times 100 = 25,000$ kgs

(2) Calculation of Joint cost

Process	Inputs	Variable costs	Fixed cost	Total cost
R	✓ 25,000	$25,000 \times 5 = 1,25,000$	42,000	1,67,000
S	✓ 22,500	$22,500 \times 4.5 = 1,01,250$	5,000	1,06,250
T	✓ 20,250	$20,250 \times 3.4 = 68,850$	4,800	73,650
				3,46,900

Raw material J = $10,000 \times 15 = ₹ 1,50,000$

Raw material K = $10,000 \times 9 = ₹ 90,000$

Raw material L = $5,000 \times 7 = ₹ 35,000$

₹ 2,75,000

Add: Processing cost (as above) ₹ 3,46,900

Total joint cost ₹ 6,21,900

(i) Statement showing apportionment of joint cost

Particulars	Product X	By-Product Z	Total
Units	14,580	3,465	
Selling price (₹)	60	30	
Sales value (₹)	✓ 8,74,800	✓ 1,09,350	9,84,150
Share of joint cost (₹ 6,21,900 in ratio of sales value)	5,52,800	69,100	6,21,900

(ii) Statement of profitability

Particulars	Product X	By-Product Z	Total
Sales value (₹)	8,74,800	1,09,350	9,84,150
(-) Share of joint cost	(5,52,800)	(69,100)	(6,21,900)
Profit	3,22,000	40,250	3,62,250

Question – 8

Mayura Chemicals Ltd. buys a particular raw material at ₹ 8 per litre. At the end of the processing in Department-1, this raw material splits-off into products X, Y and Z. Product X is sold at the split-off point, with no further processing. Products Y and Z require further processing before they can be sold. Product Y is processed in Department-2, and Product Z is processed in Department-3. Following is a summary of the costs and other related data for the year 2019-20:

Particulars	Department					
	1	2	3			
Cost of Raw Material →	₹ 4,80,000	-	-			
Direct Labour →	₹ 70,000	₹ 4,50,000	₹ 6,50,000			
Manufacturing Overhead →	₹ 48,000	₹ 2,10,000	₹ 4,50,000			
	JC 5.982	6.602	FPC 112			
Prod. (Sale + Cl. - Op.)	X (15000)	Y (15000)	Z (30000)			
Sales (litres) →	10,000	15,000	22,500			
Closing inventory (litres) →	5,000	-	7,500			
Sale price per litre (₹) →	30	64	50			

There were no opening and closing inventories of basic raw materials at the beginning as well as at the end of the year. All finished goods inventory in litres was complete as to processing. The company uses the Net-realizable value method of allocating joint costs.

You are required to prepare:

- Schedule showing the allocation of joint costs
- Calculate the cost of goods sold of each product and the cost of each item in Inventory
- A comparative statement of Gross Profit

Solution**(i) Statement of allocation of joint cost**

Particulars	Product X	Product Y	Product Z	Total
Units sold	10,000	15,000	22,500	
Add: Closing stock (A)	5,000	-	7,500	
Units Produced (B) →	15,000	15,000	30,000	
Selling price per unit (C) →	30	64	50	
Sale value of Prod. (B × C) →	4,50,000	9,60,000	15,00,000	29,10,000
Less: Additional cost (FPC) →	-	✓ 6,60,000	11,00,000	17,60,000
Net realizable value →	4,50,000	3,00,000	4,00,000	11,50,000
Share of joint cost (D)	2,34,000	✓ 1,56,000	✓ 2,08,000	5,98,000
(5,98,000 in NRV ratio)				

COP (JC + FPC) 2.342 8.162 13.082
 (+) Op. St. 2.342 × 5000 = 78000
 (-) Cl. St. 13000
 COGS 1.562 8.162 9.812
 (3.772) = 13.082 × 7500 / 30000

(ii) Statement of calculation of cost of goods sold and inventory

Particulars	Product X	Product Y	Product Z	Total
Share of joint cost	2,34,000	1,56,000	2,08,000	5,98,000
Add: Additional costs	-	6,60,000	11,00,000	17,60,000
Less: Cost of inventories	$\frac{2,34,000 \times 5,000}{15,000} =$ (78,000)	-	$\frac{13,08,000 \times 7,500}{30,000} =$ (3,27,000)	(4,05,000)
Cost of goods sold	1,56,000	8,16,000	9,81,000	19,53,000

(iii) Statement of calculation of gross profit

Particulars	Product X	Product Y	Product Z	Total
Units sold	10,000	15,000	22,500	
Selling price per unit	30	64	50	
Sales	3,00,000	9,60,000	11,25,000	23,85,000
Less: Cost of goods sold	1,56,000	8,16,000	9,81,000	19,53,000
Profit / (loss)	1,44,000	1,44,000	1,44,000	4,32,000

Question - 9

A Company produces two joint products P and Q in 70:30 ratio from basic raw materials in department A. The input output ratio of department A is 100:85. Product P can be sold at the split off stage or can be processed further at department B and sold as product AR. The input output ratio is 100:90 of department B. The department B is created to process product P only and to make it product AR.

The selling prices per kg are as under:

Product P → ₹ 85

Product Q → ₹ 290

Product AR → ₹ 115

Qty.	$80 \times \frac{85}{100} \times \frac{70}{100} = 4.76 \text{ l}$	$80 \times \frac{85}{100} \times \frac{30}{100} = 2.04 \text{ l}$
Sale	404.60	591.60
(-) S. Exp.	24.60	21.60
Netv at Split off	380.00	570.00

The production will be taken up in the next month.

Raw materials 8,00,000 kgs → $80 \times 80 = 6400$

Purchase price ₹ 80 per kg

$$SC = 6400 + 1500 = 7900$$

	Deptt. A (₹ Lacs)	Deptt. B (₹ Lacs)
Direct materials	35.00	5.00
Direct labour	30.00	9.00
Variable overhead	45.00	18.00
Fixed overheads	40.00	32.00
Total	150.00	64.00
Selling Expenses	31.86	
Product P	24.60	
Product Q	21.60	
Product AR	16.80	

Required:

- Prepare a statement showing the apportionment of joint costs if Joint Costs are apportioned in the proportion of NRV at Split-off point.
- State whether it is advisable to produce product AR or not.

Solution

(a) Input in Department A = 8,00,000 units

Output in Department A = $8,00,000 \times 85\% = 6,80,000$ units

Output of Product P = $6,80,000 \times (70/100) = 4,76,000$ units

Output of Product Q = $6,80,000 \times (30/100) = 2,04,000$ units

Statement of Joint Cost

Particulars	Amount (₹ in lakhs)
Raw Material ($8,00,000 \times 80$)	640
Direct materials	35
Direct labour	30
Variable overheads	45
Fixed overheads	40
Total Joint Cost	790

Statement of Apportionment of Joint Cost

(₹ in lakhs)

Particulars	Product P	Product Q
Sale at split off	$4.76 \times 85 = 404.60$	$2.04 \times 290 = 591.60$
Less: Selling expenses	24.60	21.60
Net realizable value at split off	380	570
Share of Joint Cost (790 in 38:57)	316	474

(b) Statement of Evaluation of Proposal

Particulars	Amount (₹ in lakhs)
Sales after further processing cost (w.n.-1)	492.66
(-) Sale at split off	404.60
Incremental sales	88.06
(-) Further processing cost	64
(+) Savings in selling expenses ($24.60 - 16.80$)	7.80
Net Benefit	31.86

Since there is net benefit, thus it is recommended to further process Product P into Product AR.

Working note – 1

Input in Department B = 4,76,000 units

Output in Department B i.e. Product AR = $4,76,000 \times 90\% = 4,28,400$ units

Sale value after further processing cost = 115×4.284 lakhs = ₹ 492.66 lakhs